

Day 8 — Data Quality Assessment

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Dataset: LG HG2 18650 Li-ion Battery Dataset | Scope: All 6 temperature folders, Mixed1 files (364,168 valid rows after PAU filter)

Part 1: Feature-by-Feature Data Quality Table

The following table was produced by auditing all 364,168 valid TABLE rows across the six Mixed1 files (one per temperature). All statistics are computed on actual data without any cleaning applied.

Feature	Type	Valid Rows	Missing	Min	Max	Mean	Std	Quality	Notes
Voltage (V)	Float	364,168	0 (0%)	2.795	4.202	3.692	0.284	Good	No values outside physical bounds (2.5–4.3V)
Current (A)	Float	364,168	0 (0%)	−16.89	+6.00	−1.35	2.47	Good	Negative=discharge, Positive=charge. No values outside ±20A
Temperature (°C)	Float	364,168	0 (0%)	−19.98	+40.17	Varies	Varies	Note	Self-heating under load: −10°C folder shows actual range −10.1 to −6.3°C
Capacity (Ah)	Float	364,168	0 (0%)	−2.590	0.000	−1.075	0.712	Note	Always ≤ 0 in discharge files. 218 rows at exactly 0 (segment start points)
WhAccu (Wh)	Float	364,168	0 (0%)	−9.6	0.000	−4.01	2.71	Good	Cumulative energy. Mirrors Capacity pattern. Consistent with V×I×t.
Time Stamp	String→datetime	364,168	0 (0%)	—	—	—	—	Good	Timestamps monotonically increasing within all 6 files. Zero non-monotonic rows detected.
Status	Categorical	400,174 total	0 (0%)	—	—	—	—	Action	36,006 PAU rows (9.0% of total). Must be filtered before use.
Step	Integer	364,168	0 (0%)	1	Varies	—	—	Good	Test step index. Consistent within files. Useful as a segment marker.
Cycle	Integer	364,168	0 (0%)	0	Varies	—	—	Good	Sub-cycle counter. Can be used to identify individual drive cycles within a session.
Prog Time	String→timedelta	364,168	0 (0%)	—	—	—	—	Note	Format HH:MM:SS.ms. Useful for time_elapsed feature. Must be parsed carefully.
Procedure	String	364,168	0 (0%)	—	—	—	—	Good	Test program name. Same value within a file. Informational only.

Part 2: Detailed Quality Findings

2.1 Missing Values

After dropping the unit descriptor row (row index 0 after the column header) and filtering out PAU rows, zero missing values were found in any sensor column across all 6 Mixed1 files. This is exceptional data quality for a real experimental dataset.

Column	NaN in raw file	Cause	After fix
All sensor columns	1 per file (6 total)	Unit descriptor row (V, A, °C, Ah) immediately after column header row	0 NaN after drop(index=0)

Column	NaN in raw file	Cause	After fix
All other columns	0	Not applicable	0 NaN — no action needed

2.2 Invalid Ranges and Outliers

Physical validity ranges for the LG HG2 18650 cell are: Voltage 2.5V to 4.25V, Current –20A to +10A, Temperature –30°C to 60°C. All observed values fall within these bounds after PAU filtering.

Feature	Physical Bounds	Observed Range	Out-of-bounds	IQR Outliers	Verdict
Voltage	2.5V to 4.25V	2.795V to 4.202V	0	5,373 (1.48%)	No invalid values. IQR outliers are valid extreme SoC readings.
Current	–20A to +10A	–16.89A to +6.00A	0	Not applicable — bimodal distribution	No invalid values. Bimodal shape expected (charge vs discharge).
Temperature	–30°C to 60°C	–19.98°C to +40.17°C	0	Not applicable	No invalid values. Self-heating under load is expected and valid.
Capacity	–3.5Ah to 0Ah	–2.590Ah to 0.000Ah	0	Not applicable	No positive values found. 218 zero rows are valid segment starts.

Note on IQR outliers for Voltage: The 5,373 IQR outliers (1.48%) represent valid extreme readings near the lower voltage cutoff (around 2.8V at –20°C) and at full charge (4.20V). These are NOT errors — they are physically expected readings at the edges of the SoC range and must NOT be removed.

2.3 Duplicates

Two types of apparent duplicates were identified. Neither represents genuine data errors.

Duplicate Type	Count	% of Valid Rows	Cause	Action
Exact (V, I, T, Cap match)	21,654	5.9%	Tester holding steady-state values during rest periods between drive cycles. All four sensor values unchanged for multiple timesteps.	Do NOT remove. These are valid rest-state readings critical for SoC continuity.
Near-duplicate (V, I, T match)	74,253	20.4%	In addition to the above, capacity increments by a tiny amount each step during rest (coulomb counting continues). V, I, T are the same but capacity differs slightly.	Do NOT remove. The Capacity column distinguishes these rows during SoC derivation.

2.4 PAU Status Rows

PAU (tester paused) rows represent periods when the Digatron tester is in a hold state between test steps. During PAU, the cell is at rest and sensor readings may not reflect active battery state. However, PAU readings are not random noise — they are consistent rest-state values.

Temperature	Total Rows	PAU Rows	TABLE Rows	PAU %	Notes
40°C	75,342	6,001	69,341	7.97%	Consistent 6,001 PAU rows across all temps
25°C	77,228	6,001	71,227	7.77%	Same pause structure — fixed by test program
10°C	72,330	6,001	66,329	8.30%	
0°C	67,313	6,001	61,312	8.92%	PAU % rises as sessions get shorter at cold temps
–10°C	62,623	6,001	56,622	9.58%	
–20°C	45,344	6,001	39,343	13.23%	Highest PAU% — battery sessions very short at –20°C

2.5 Temperature Self-Heating

A key finding from the temperature audit is that the measured temperature inside each file does not match the folder's nominal temperature during active discharge. This is because the battery cell self-heats under load. The effect is larger at colder temperatures.

Folder	Nominal Temp	Measured Min	Measured Max	Self-heat range	Impact on Modelling
40degC	40°C	38.38°C	40.17°C	+1.79°C	Minor. Use measured temperature column.
25degC	25°C	23.56°C	25.87°C	+2.31°C	Minor. Use measured temperature column.
10degC	10°C	8.83°C	12.62°C	+3.79°C	Moderate. Temperature varies significantly within file.
0degC	0°C	–0.74°C	+2.94°C	+3.68°C	Significant. Cell warms from near-zero to +3°C during discharge.
n10degC	–10°C	–10.10°C	–6.31°C	+3.79°C	Significant. Use measured temperature, not folder name.

Folder	Nominal Temp	Measured Min	Measured Max	Self-heat range	Impact on Modelling
n20degC	-20°C	-19.98°C	-15.25°C	+4.73°C	Largest self-heating. Never use -20 as a fixed feature value.

2.6 Target Coverage

The SoC target (derived from Capacity column) and its coverage across temperature conditions is assessed below. Note that usable SoC span decreases substantially at lower temperatures.

Temperature	Capacity Range (Ah)	Approx SoC Span (%)	Implication
25°C	-2.590 to 0	86.3%	Widest SoC coverage. Model sees SoC from ~14% to ~100%.
40°C	-2.526 to 0	84.2%	Similar to 25°C. Slight capacity reduction at high temp.
10°C	-2.480 to 0	82.7%	Mild reduction. Still good coverage.
0°C	-2.350 to 0	78.3%	Noticeable capacity reduction. Lower SoC range narrower.
-10°C	-2.130 to 0	71.0%	Significant reduction. Model may not see low-SoC behavior well.
-20°C	-1.588 to 0	52.9%	Severe reduction. Only covers upper half of SoC range. Evaluate separately.

2.7 Leakage Check

Temporal leakage is the most critical risk for this time-series dataset. The following checks were performed:

Timestamp monotonicity: All 6 Mixed1 files have strictly increasing timestamps. Zero non-monotonic timestamp rows detected. This confirms no internal file ordering issues.

Cross-file leakage risk: The dataset has no cross-file references. Each file is independently collected. However, if rows from different files are mixed before splitting, temporal order is violated. The planned file-level split (not row-level) prevents this.

Feature look-ahead: Rolling window features (planned for Day 9) must be computed within each file separately. Computing rolling windows across file boundaries would introduce future information into past predictions.

Scaler leakage: StandardScaler must be fit on training files only. Fitting on the full dataset before splitting is a common leakage error that inflates test performance.

Part 3: Proposed Cleaning Rules with Justification

The following cleaning rules are proposed based on the quality assessment above. Rules are ordered by priority. Each rule includes the justification, the code approach, and whether it is reversible.

Rule ID	Rule	Justification	Implementation	Reversible
R1	Drop unit descriptor row (index 0 after header)	The first row after the column header contains unit strings (V, A, °C, Ah) not numeric data. Causes all sensor columns to parse as object dtype.	<code>df.drop(index=0).reset_index(drop=True)</code> after <code>pd.read_csv()</code>	Yes — raw file unchanged
R2	Filter rows where Status = PAU	PAU rows represent tester pause periods. Sensor readings during PAU are rest-state values not representative of active driving or charging behavior. Including them inflates rest-state SoC clusters.	<code>df = df[df['Status'] != 'TABLE']</code>	Yes — applied on load, raw unchanged
R3	Force numeric dtype on all sensor columns	Mixed-type columns (object dtype) cause silent failures in arithmetic. The unit row causes dtype detection to fail for all sensor columns.	<code>pd.to_numeric(df[col], errors='coerce')</code> for each sensor column	Yes — in-memory only
R4	Do NOT remove IQR outliers in Voltage	The 5,373 IQR outliers represent valid readings near the cutoff voltage (2.8V at -20°C) and at full charge (4.2V). Removing them would eliminate critical SoC range data from the model.	No removal. Use physical bounds (2.5–4.3V) as the only filter.	Not applicable
R5	Do NOT remove near-duplicate rows	Near-duplicates (V,I,T match) are valid rest-state readings. They provide information about battery behavior at steady state and must be preserved for accurate SoC derivation via Capacity column.	No deduplication. Drop only if exact (V,I,T,Cap,Time) match — genuine tester error.	Not applicable
R6	Use measured Temperature column — not folder name	Self-heating under load causes the actual cell temperature to differ from the nominal folder temperature by up to 4.73°C at -20°C. Using the folder name as a temperature feature would be incorrect.	Always use <code>df['Temperature']</code> as input feature. Never hardcode folder nominal temperature.	Not applicable

Rule ID	Rule	Justification	Implementation	Reversible
R7	Exclude Charge files with fewer than 500 rows	Several Charge files (e.g., Charge1 at 25°C) contain only 131 rows — these represent only the CV tail of a charging session. Too few rows to reliably derive Remaining Charging Time labels.	<code>len(df) >= 500</code> check before adding to Charge file list. Document excluded files.	Yes — simple threshold filter
R8	Split by file — never by row	Battery data is time-series within each file. Row-level splitting destroys temporal continuity, allows future sensor values into the training set (leakage), and inflates test performance.	Assign entire files to train/val/test sets before any processing. Apply scaler after split.	Not applicable — architectural decision
R9	Fit StandardScaler on training files only	Fitting on the full dataset before splitting leaks test set statistics into the scaler. This inflates test performance by centering and scaling using distribution knowledge from the test set.	<code>scaler.fit(X_train)</code> then <code>scaler.transform(X_val)</code> and <code>scaler.transform(X_test)</code>	Not applicable — must be enforced
R10	Compute rolling features within each file	Rolling averages computed across file boundaries introduce data from one session into another. The end of one charging session should not influence the start of a different drive cycle.	Apply <code>rolling()</code> within each file's DataFrame separately before concatenation.	Not applicable — architectural decision

3.1 Cleaning Rule Priority Order

Rules should be applied in the following sequence during data loading:

Step 1: Load CSV, skip 28 header rows (R3)

Step 2: Drop unit row at index 0 (R1)

Step 3: Force numeric dtypes on all sensor columns (R3)

Step 4: Filter Status = PAU rows (R2)

Step 5: Apply physical bounds check — flag but do not remove outliers (R4)

Step 6: Use measured Temperature column as feature, not folder name (R6)

Step 7: Assign files to train/val/test (R8)

Step 8: Compute rolling features within each file (R10)

Step 9: Fit scaler on training data only, apply to val and test (R9)

Rules R4, R5, R7, R8, R9, and R10 are constraints rather than cleaning steps. They define what NOT to do. Violating any of these six would compromise model validity.

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